

THE LARGEST CYCLIC 3-(31, 5, 1) PACKING HAS TWELVE BASE-BLOCK ORBITS

ZIJIAN ZENG

ABSTRACT. We determine the largest number of full translation orbits in a cyclic 3-(31, 5, 1) packing. A twelve-orbit construction is verified by developing all 372 blocks and checking all 3,720 contained triples. For the upper bound, the problem is reconstructed as an exact set packing on 145 triple-translation resources and 4,761 internally valid block-translation classes. Multiplication by the units of $\mathbb{Z}_{31}$ partitions these classes into 162 orbits. A definition-level integer enumerator rejects a packing of size thirteen in every fixed-orbit branch. A separate fail-closed verifier reconstructs the finite instance and the complete multiplier cover. Thus the maximum number of base-block orbits is exactly twelve.

1. INTRODUCTION

Let $G = \mathbb{Z}_{31}$. We seek as many translation classes of five-subsets of G as possible, subject to the condition that no three-subset occurs in more than one developed block. This is equivalently a one-dimensional optical orthogonal code with weight five and both correlation bounds equal to two; the general OOC framework was introduced by Chung, Salehi, and Wei [3]. It is important not to confuse this problem with the classified $(v, 5, 2, 1)$ case, whose cross-correlation bound is one [1].

The source problem was recorded with the interval $9 \leq M \leq 13$ [4]. We independently reconstruct that finite model, improve the lower bound to twelve, and close the remaining one-bit gap. Our result is the following.

Theorem 1.1. *The maximum number M of base-block orbits in a cyclic 3-(31, 5, 1) packing is*

$$M = 12.$$

The proof is computation-assisted but exact. All feasibility decisions use integer set operations and bit sets. Floating-point elapsed times and failed or timed-out optimizer runs play no role.

2. EXACT ORBIT FORMULATION

For a subset $S \subseteq G$, write $[S] = \{S + t : t \in G\}$. Every nonempty proper subset of G has a full translation orbit: if $S + t = S$ for nonzero t , then t

generates the prime-order group and S is either empty or all of G . It follows that there are

$$\binom{31}{3}/31 = 145 \quad \text{and} \quad \binom{31}{5}/31 = 5,481$$

translation classes of triples and five-subsets, respectively.

For a five-subset B , let $\rho(B)$ be the multiset of translation classes of its ten triples. The orbit $[B]$ is internally valid precisely when the ten entries of $\rho(B)$ are distinct. Two internally valid block classes are compatible precisely when their ten resource sets are disjoint. Exact enumeration leaves 4,761 internally valid classes. Consequently the original problem is the finite set packing

$$\max \sum_{i=1}^{4761} x_i, \quad \sum_{i:r \in \rho(B_i)} x_i \leq 1 \quad (r = 1, \dots, 145), \quad x_i \in \{0, 1\}.$$

The usual packing framework and Johnson–Schönheim bounds are discussed, in greater generality, by Bailey and Burgess [2].

3. A TWELVE-ORBIT PACKING

The following base blocks are written with their least element equal to zero:

$$\begin{aligned} &\{0, 1, 4, 12, 23\}, \quad \{0, 1, 5, 8, 15\}, \quad \{0, 1, 6, 7, 19\}, \\ &\{0, 1, 9, 18, 24\}, \quad \{0, 1, 11, 14, 17\}, \quad \{0, 2, 4, 15, 19\}, \\ &\{0, 2, 6, 24, 27\}, \quad \{0, 2, 7, 11, 18\}, \quad \{0, 2, 12, 21, 26\}, \\ &\{0, 2, 16, 22, 28\}, \quad \{0, 3, 8, 14, 24\}, \quad \{0, 4, 8, 13, 21\}. \end{aligned}$$

Proposition 3.1. *The full translation development of these blocks is a 3-(31, 5, 1) packing.*

Proof. The strict verifier expands every block by all 31 translations. It obtains 372 distinct developed blocks and 3,720 distinct contained triples. Since each of the $372 \binom{5}{3} = 3,720$ triple incidences has a distinct owner, no triple is repeated. This proves $M \geq 12$. $\square$

4. STRUCTURAL UPPER BOUNDS

For $d \in \{1, \dots, 15\}$, an unordered pair has distance d when its difference is congruent to $\pm d \pmod{31}$. Let $w_d(B)$ be the number of pairs of distance d in B , and set $r_d = \sum_B w_d(B)$ over selected base blocks. Every fixed pair of distance d lies in exactly r_d developed blocks. Each such block uses three of the 29 triples through that pair, and distinct packing blocks use disjoint third points. Therefore

$$r_d \leq \left\lfloor \frac{29}{3} \right\rfloor = 9.$$

Because every base block has ten pairs,

$$10M = \sum_{d=1}^{15} r_d \leq 135,$$

and hence $M \leq 13$.

For completeness, if $M = 13$ and $l_d = 29 - 3r_d$, then

$$l_d = 2 + 3e_d, \quad e_d \geq 0, \quad \sum_{d=1}^{15} e_d = 5.$$

This leave condition is used as an independently checked structural filter, but the global exclusion below does not rely on a floating-point relaxation.

5. EXACT EXCLUSION OF THIRTEEN ORBITS

Multiplication by $a \in \mathbb{F}_{31}^\times$ permutes block and triple translation classes and preserves feasibility. Direct reconstruction of this action partitions the 4,761 internally valid block classes into

156 orbits of size 30, 5 orbits of size 15, 1 orbit of size 6.

Thus there are 162 multiplier orbits in all.

Choose the least-numbered variable in each multiplier orbit. If a thirteen orbit packing existed, choose any one of its blocks and multiply the entire packing so that this block becomes the chosen representative of its orbit. The other twelve blocks would then form a 12-clique in the representative's compatibility neighborhood. It is therefore complete to reject these 162 fixed-neighborhood clique instances.

Lemma 5.1. *The fixed-neighborhood enumerator is complete: if it reports UNSAT_BY_COMPLETE_BRANCHING, then its compatibility neighborhood has no clique of size twelve.*

Proof. The program rebuilds the 145 resources and 4,761 columns from the definitions, then creates an edge exactly for disjoint resource sets. At a recursive node, it greedily partitions the candidate graph into independent color classes. The number of colors is an upper bound on the size of a clique in that candidate set. Vertices are processed in reverse nondecreasing color order, so a failed prefix color bound safely rejects all unprocessed branches.

Otherwise the recursion considers each remaining vertex. Its child set is the intersection of the remaining prefix with its exact adjacency set. The additional distance-pair filter rejects a vertex only when some accumulated r_d would exceed nine, which is impossible in a packing by the preceding section. Hence the include and exclude branches are exhaustive, all pruning bounds are necessary, and the finite candidate set strictly decreases down every recursive edge. $\square$

Proposition 5.2. *No cyclic 3-(31, 5, 1) packing has thirteen base-block orbits.*

Proof. The exact enumerator rejects a neighbor clique of size twelve for each of the 162 multiplier representatives. The multiplier orbits are disjoint and cover all 4,761 block variables, so the preceding symmetry argument covers every hypothetical thirteen-orbit packing. The complete run explored 68,377,851,660 deterministic branching nodes; individual branches ranged from 251,805,613 to 636,752,470 nodes. These counts are reproducibility checks, not logical assumptions. The global fail-closed verifier independently rebuilds the finite instance and multiplier partition, checks every branch manifest, and reports `VERIFIED_GLOBAL_TARGET13_UNSAT`. $\square$

Theorem 1.1 now follows from Propositions 3.1 and 5.2.

6. CERTIFICATES, AUDITING, AND REPRODUCIBILITY

The principal locked artifacts are listed in Table 1.

Artifact	SHA-256
Exact instance	f43bb843344058df6794c5c86e314f547292a65fa43183d2678ef178e5e4034d
12-block certificate	23dc758771e6dce03b943a48aab50077a5ee00c36e24cde02f74aa2468f8cf45
Global target-13 manifest	ae259e64de1793421a7e5e4bd525fec405133f3f277074a222badf6ac5d522b3
Enumerator source	ae549763147bd579353d3b91fcbad4e9afb4e2e404c97f915115d8b6e84458c9
Enumerator binary	2b0cb86d49c2ce6add774dedf98796b706d72cd0d2266b5a7aa9ce653413d30f

TABLE 1. Hashes of the central serialized proof artifacts.

From the project root, the short verification commands are

```
python3 code/verify_packing_certificate.py \
  certificates/packing12.json --expected-blocks 12
python3 code/verify_global_fixed_cover.py
python3 tests/test_global_fixed_cover.py
python3 tests/test_fixed_clique_certificate.py
```

The optional `-rebuild-rerun` mode compiles the locked C++ source in a fresh temporary directory and repeats the exact search. Clean rebuild replays were performed for the minimum-node branch and the unique size-six multiplier orbit. The positive, fixed-branch, global-cover, SAT/PB encoding, leave, and symmetry tests all include adversarial rejection checks. Two auxiliary LRAT refutations for one leave-pattern orbit were also checked with the official `lrat-check`; they are cross-checks rather than the global upper bound.

No proof assistant was used. The upper bound is a certified finite result whose checker is conventional source code, not a Lean/Coq formalization.

COMPUTATIONAL AND AI DISCLOSURE

OpenAI Codex was used as a computational research assistant for model construction, program execution, verification, literature screening, and manuscript preparation. The sole author selected and audited the mathematical claims. Discovery outputs, optimizer timeouts, floating-point bounds, and interrupted runs were excluded from the proof. A post-result search through the documented databases found no earlier resolution of this exact parameter; that novelty statement is necessarily limited to the searched and indexed literature.

REFERENCES

1. Tsonka Baicheva and Svetlana Topalova, *Optimal $(v, 5, 2, 1)$ optical orthogonal codes of small v* , Applicable Algebra in Engineering, Communication and Computing **24** (2013), no. 3-4, 165–177.
2. Robert F. Bailey and Andrea C. Burgess, *Generalized packing designs*, Discrete Mathematics **313** (2013), no. 11, 1167–1190.
3. F.R.K. Chung, J.A. Salehi, and V.K. Wei, *Optical orthogonal codes: design, analysis and applications*, IEEE Transactions on Information Theory **35** (1989), no. 3, 595–604.
4. TheoremDB contributors, *Largest cyclic 3-(31, 5, 1) packing*, <https://www.theoremdb.org/statements/cyclic-315-packing-31/>, 2026, Research-memory snapshot of July 25, 2026.

INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR, 56000, MALAYSIA

Email address: `zijianzeng@foxmail.com`

Email address: `1002266693@ucsiuniversity.edu.my`